radioactivity of, 2, 3
species in water, 767f, 769
SSNTD for, 243f, 245
uses of, 418
Uranium once-through (UOT), 687f, 706
Uranium ores, 414
gamma-radiation from, 388
Uranium radioelement series, 4, 382–386
Uranium-plutonium (U-Pu) fuel cycle, 706–708, 707t

V

Valley of β-stability, 77–79, 77f–78f
for promethium, 79, 80f
for technetium, 80–82, 81f
Van de Graaff accelerators (VdG), 498f, 499–502
VdG, *see* Van de Graaff accelerators
Vector quantity, 131–132
Velocity
average particle, 44
of charged particles, 174f, 175
of electrons, 22–23
gravitational pull and, 343
most probable, 45
radial, 345–346
in recoilless absorption, 197–198, 197f
Very large whole-body doses, 446
Vibrational energy, 46
in bimolecular reaction, 50–51, 50f
in deformed nuclei model, 144–145, 145f
Vibrational quantum number, 46
Virgo supercluster, 340
V_n, *see* Nuclei volume
Vodo-Vodyanoi Energetichesky reactors (VVER), 657
overview of, 659
Void coefficient, 630
Voltage-sensitive preamplifiers, 275, 275f
Volume energy, in semiempirical mass equation, 76–77
VVER, *see* Vodo-Vodyanoi Energetichesky reactors
VVR, history of, 12

W

W^-
strength and range of, 17t
as weak force mediators, 29
W^+
strength and range of, 17t
as weak force mediators, 29
Waste repositories
natural analogues for, 776–777
Oklo reactor and, 777–778
performance assessment of, 778–786
canister dissolution, 772f, 774f, 779f, 780–782, 781f
migration from, 781f, 782–786, 784t, 785t
release from bitumen and concrete, 782
release scenarios, 772f, 779–780
speciation calculations for, 767f, 769–776, 775t
in solution, 771f, 773–776
Waste streams
in isotope separation, 52–53, 53f
from reprocessing, 723–728, 724t
environmental releases of, 727–728, 727t
gaseous, 723–725
liquid, 725, 726t
organic, 726
solid, 724t, 726–727
Water
electromagnetic radiation in, 213f, 233
neutron moderation with, 606–607, 606t
quenching effect of, 272, 273f
radiation effects on, 219–222, 222f
radiation range in, 172, 172t
radiolysis of, 209–210, 220–221, 220f, 221t
spur reactions in, 220–221, 221t
Wavelength shifters, 270
Waves
particles and, 19–20
photons as, 20
Weak interaction force
reaction time of, 18–19
strength and exchange particles of, 17t, 18
strong nuclear force compared with, 18–19
Wear, radioactive tracers used in studies of, 588
Westinghouse design, of PWR, 625, 626f
White dwarf, 354
Whole-body counters, 283
Whole-body counting, 489

Wideröe linear accelerator, 499, 499f
Wigner effect, 217–218, 218f

X

Xenon poisoning, 616
Xenon transient, 616
X-ray
  discovery of, 1–2
  emission of, 106
  PIXE, 192, 193f
  tracks with, 213
  in water, 213f, 233
X-ray fluorescence analysis, 194–196, 195f

Z

Z, *see* Atomic number, Protons
Z^0
  strength and range of, 17t
  as weak force mediators, 29
ZAMS, *see* Zero Age Main Sequence
Zeeman effect, 149–151, 150f
Zero Age Main Sequence (ZAMS), 344–345, 345f
Zero point vibrational frequency, 46
Zinc, in water cooled reactors, 668
Zircaloy, 689